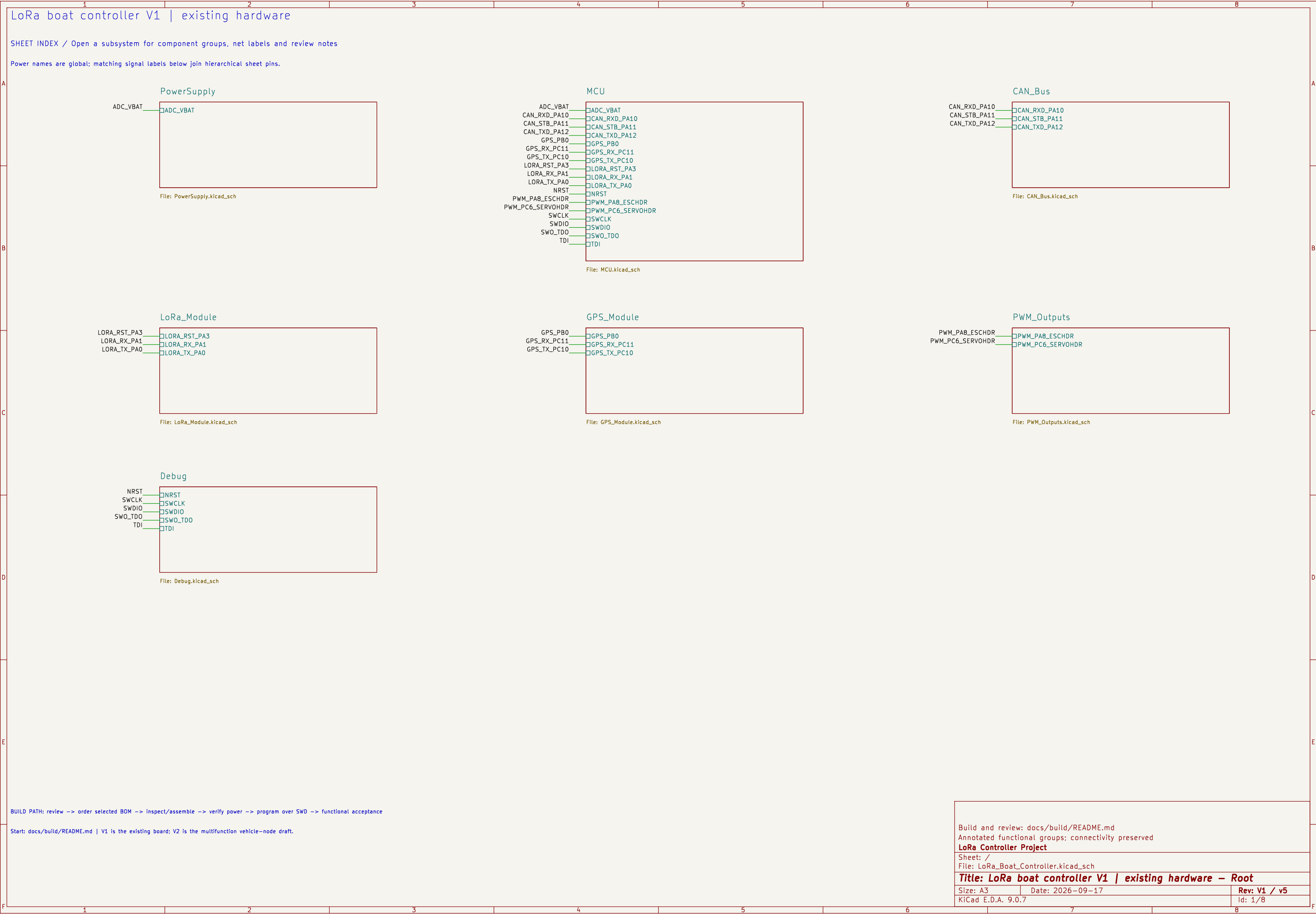
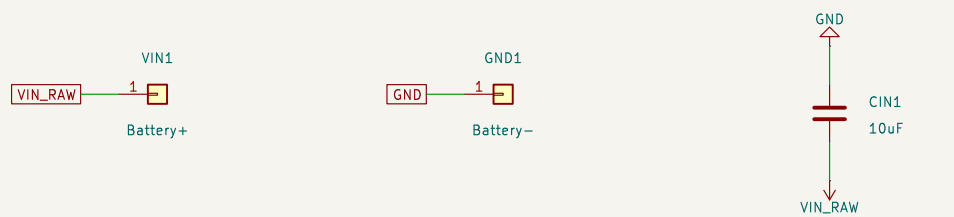




V1 / POWERSUPPLY

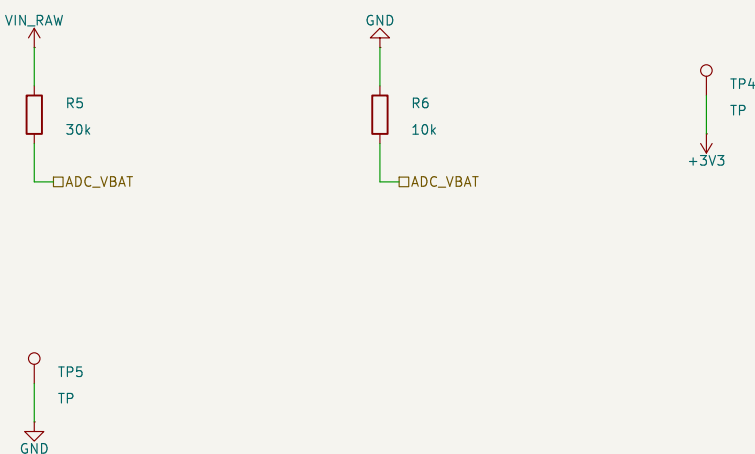
Functional groups / electrical pin numbers / named-net connections

01 Battery entry



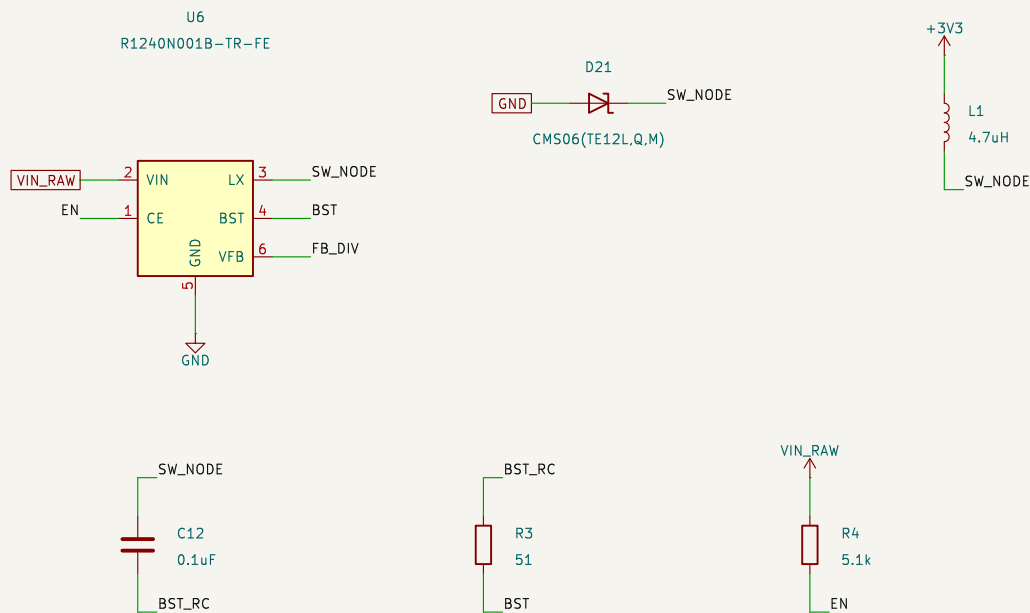
Existing V1 solder pads. No fuse, reverse protection or transient clamp.

04 Battery sensing and test access



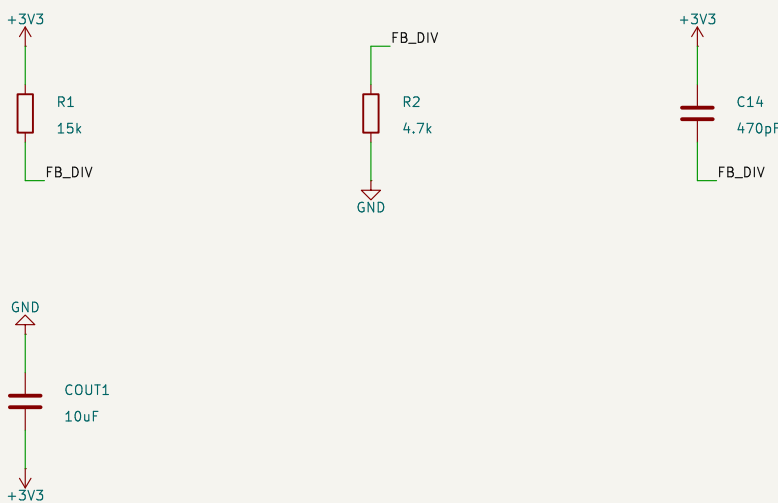
Divider is 4:1; 3.3 V ADC full scale corresponds to 13.2 V. Legacy firmware does not sample it.

02 Logic regulator and switching loop

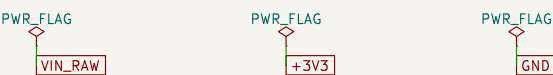


R1240N asynchronous buck. Preserve the original catch diode and bootstrap network.

03 Feedback and output reservoir



Nominal output: $0.8 \times (1 + 15k / 4.7k) = 3.35$ V. Values describe V1, not a new qualification.



Read named nets within each group; matching labels are electrically connected. Cross-sheet signals use hierarchical ports.

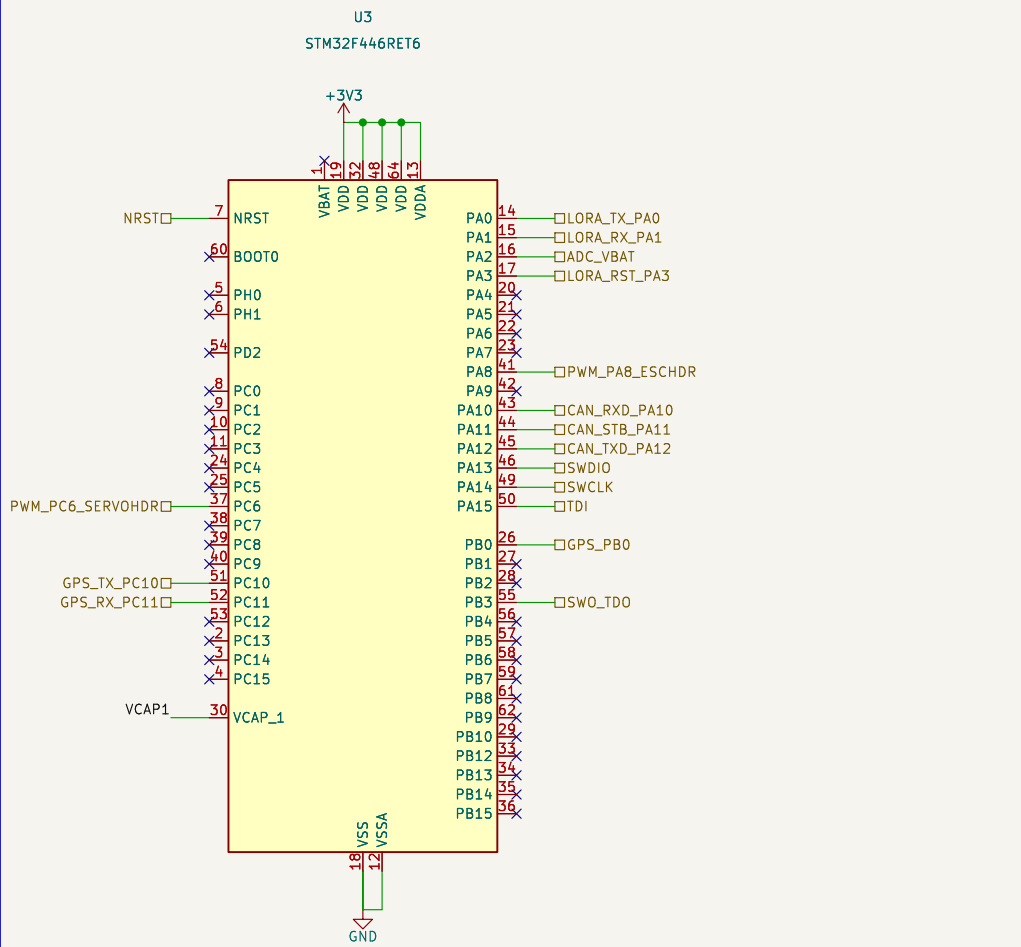
V1 preserves original defects. V2 is a review draft. Bench values and firmware behavior require validation.

Build and review: docs/build/README.md
Annotated functional groups; connectivity preserved
LoRa Controller Project
Sheet: /PowerSupply/
File: PowerSupply.kicad_sch

Title: LoRa Autonomous Boat Controller V1 - PowerSupply

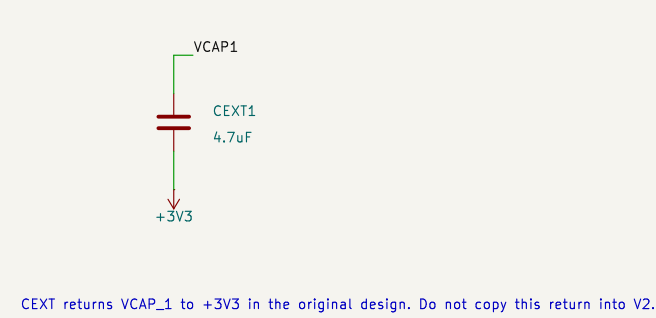
Size: A2	Date: 2026-09-17	Rev: V1 / v5
KiCad E.D.A. 9.0.7		Id: 2/8

01 MCU signal map

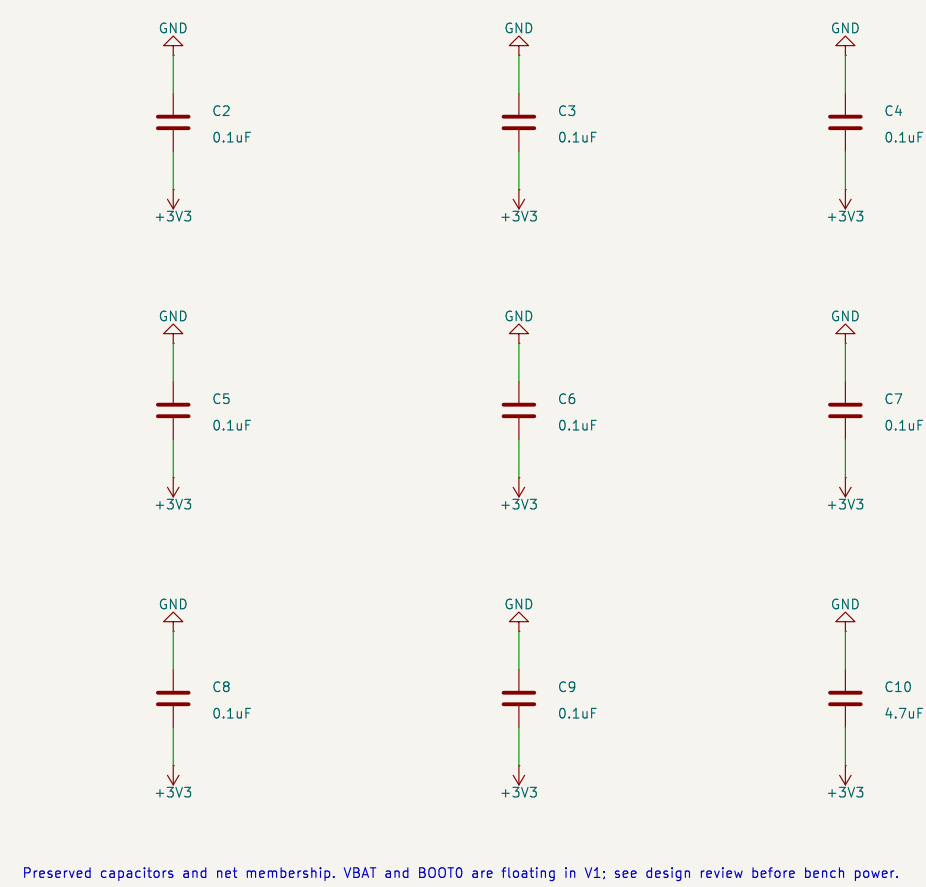


STM32F446RET6 / LQFP-64. PA8 reaches ESC-labelled header; PC6 reaches servo-labelled header. Firmware roles are reversed.

02 Core supply – preserved defect



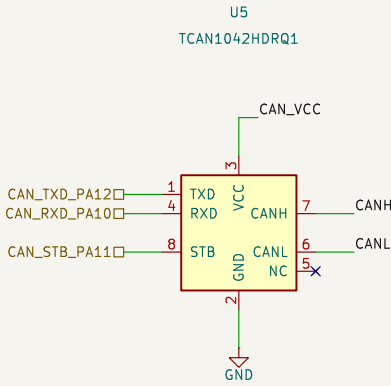
03 Logic supply decoupling



Read named nets within each group: matching labels are electrically connected. Cross-sheet signals use hierarchical ports.

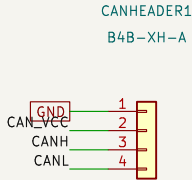
V1 preserves original defects. V2 is a review draft. Bench values and firmware behavior require validation.

01 Transceiver



VCC is external 4.5–5.5 V. RXD reaches PA10, which has no CAN receive function; V1 CAN is not operational.

02 External CAN port



Pin numbers are electrical numbering, not a cable–side view. Confirm pin 1 and supply polarity on the real board.



Read named nets within each group; matching labels are electrically connected. Cross-sheet signals use hierarchical ports.

V1 preserves original defects. V2 is a review draft. Bench values and firmware behavior require validation.

Build and review: docs/build/README.md
Annotated functional groups; connectivity preserved
LoRa Controller Project
Sheet: /CAN_Bus/
File: CAN_Bus.kicad_sch

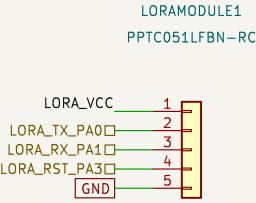
Title: LoRa Autonomous Boat Controller V1 – CAN_Bus

Size: A3	Date: 2026–09–17	Rev: V1 / v5
KiCad E.D.A. 9.0.7		Id: 4/8

V1 / LORA MODULE

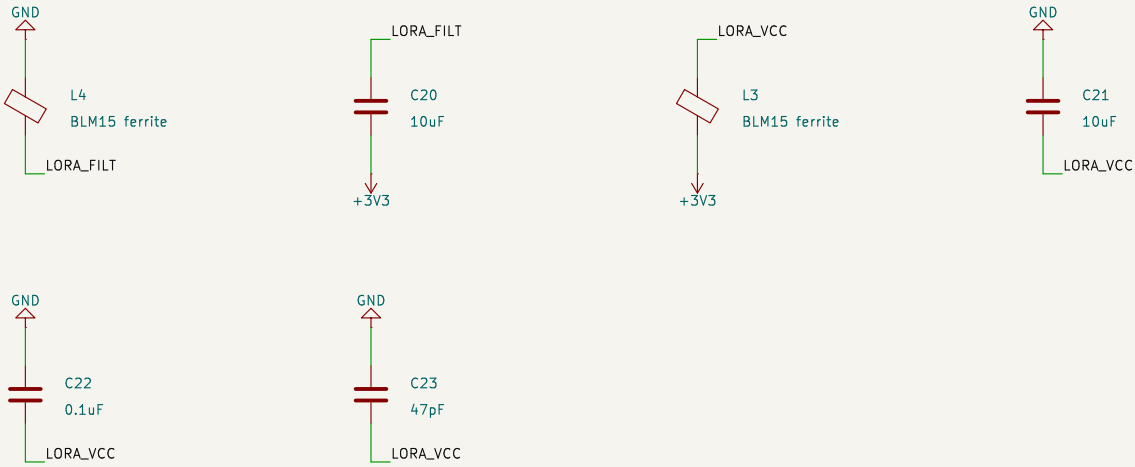
Functional groups / electrical pin numbers / named-net connections

01 UART radio socket



Identify the fitted radio and capture AT replies before selecting SF/BW or assuming reset-pin behavior.

02 Supply filter



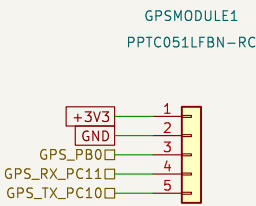
Preserved defect: C20 returns to +3V3. C23 value/MPN conflict remains in the BOM; resolve the fitted value.

Read named nets within each group; matching labels are electrically connected. Cross-sheet signals use hierarchical ports.
V1 preserves original defects. V2 is a review draft. Bench values and firmware behavior require validation.

Build and review: docs/build/README.md		
Annotated functional groups; connectivity preserved		
LoRa Controller Project		
Sheet: /LoRa_Module/		
File: LoRa_Module.kicad_sch		
Title: LoRa Autonomous Boat Controller V1 – LoRa_Module		
Size: A3	Date: 2026-09-17	Rev: V1 / v5
KiCad E.D.A. 9.0.7		Id: 5/8

Functional groups / electrical pin numbers / named-net connections

01 GNSS serial interface



PC11 is GNSS-to-MCU data; PC10 is MCU-to-GNSS data. PB0 on pin 3 is driven low by legacy firmware; confirm the module pin function.

Read named nets within each group; matching labels are electrically connected. Cross-sheet signals use hierarchical ports.

V1 preserves original defects. V2 is a review draft. Bench values and firmware behavior require validation.

Build and review: docs/build/README.md
Annotated functional groups; connectivity preserved
LoRa Controller Project
Sheet: /GPS_Module/
File: GPS_Module.kicad_sch

Title: LoRa Autonomous Boat Controller V1 – GPS_Module

Size: A3	Date: 2026-09-17	Rev: V1 / v5
KiCad E.D.A. 9.0.7	Id: 6/8	

Functional groups / electrical pin numbers / named-net connections

01 ESC-labelled connector / legacy rudder



PA8 / TIM1. Confirm physical harness before changing firmware. Pin 1 is +3V3, not a BEC input.

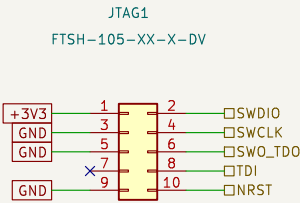
02 Servo-labelled connector / legacy thrust



PC6 / TIM3. Legacy thrust output starts at 1000 us in C; CubeMX configuration differs. Disconnect propulsion during first tests.

Read named nets within each group; matching labels are electrically connected. Cross-sheet signals use hierarchical ports.
V1 preserves original defects. V2 is a review draft. Bench values and firmware behavior require validation.

01 Cortex debug access



3.3 V target reference, SWDIO, SWCLK, NRST and ground. Probe VTref is a sense input; avoid competing power supplies. See docs/build/FLASHING_V1.md.

Read named nets within each group; matching labels are electrically connected. Cross-sheet signals use hierarchical ports.

V1 preserves original defects. V2 is a review draft. Bench values and firmware behavior require validation.

Build and review: docs/build/README.md
Annotated functional groups; connectivity preserved
LoRa Controller Project
Sheet: /Debug/
File: Debug.kicad_sch

Title: LoRa Autonomous Boat Controller V1 – Debug

Size: A3	Date: 2026-09-17	Rev: V1 / v5
KiCad E.D.A. 9.0.7	Id: 8/8	